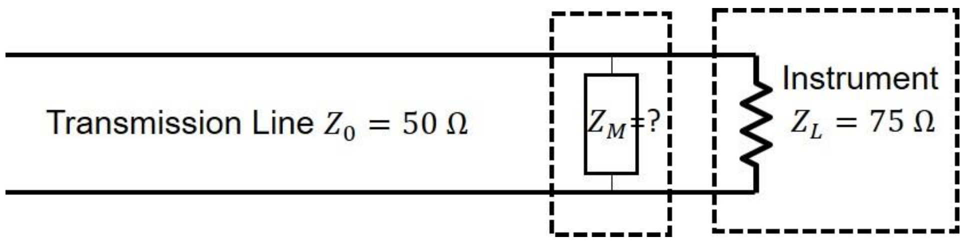
PHYS0525 Analog and Digital Electronics – Spring 2026

Assignment 4

Monday, Feb 16, 2026

1) Suppose you have a length of 50 Ω cable and want to connect it to an instrument with a 75 Ω input impedance. What circuit element (labeled Z_m in the figure below) could you add just prior to the instrument to impedance match this connection (i.e. have no back reflections into your 50 Ω cable).

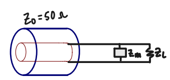


$\Rightarrow \frac{50(Z_L+Z_m)}{Z_L} = Z_m \Rightarrow \frac{50(75+Z_m)}{75} = Z_m$

$\rightarrow \frac{3750+50Z_m}{75} = Z_m \Rightarrow 75Z_m = 3750 + 50Z_m$

$25Z_m = 3750$

$Z_m = \frac{3750}{25} = 125\ \Omega$



what $Z_m \parallel$ w/ $Z_L = Z_0$?: $50\ \Omega = \frac{1}{Z_M} + \frac{1}{Z_L}$

$\hookrightarrow 50\ \Omega = \frac{Z_L Z_M}{Z_L+Z_M} \Rightarrow 50(Z_L+Z_m) = Z_L + Z_m$

* Z_{mL} must equal Z_0 to have no back reflections

2) A wave of frequency $\omega = 2\pi \times 1$ MHz and amplitude $V_0 = 1$ V is traveling down a 50 Ω cable. The cable is terminated with a capacitor with a series combination of a capacitor $C = 10\ \mu$F and a resistor $R = 25\ \Omega$.

a) What is the amplitude and phase of the voltage of the reflected wave?

b) What is the amplitude and phase of the current of the reflected wave?

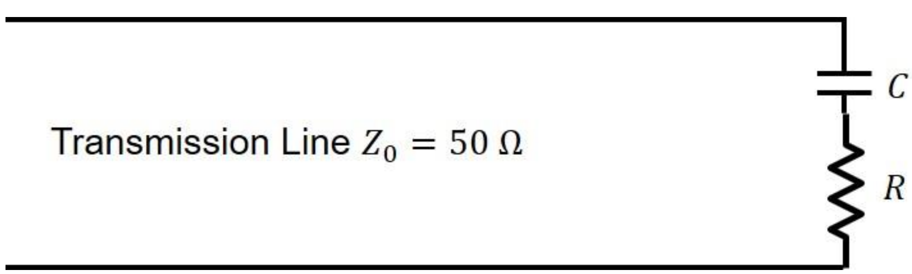
c) What is the amplitude of voltage across the resistor?

d) Compute the average power in the incident wave, the average power in the reflected wave, and the average power dissipated by the resistor.

$Z_L = \sqrt{Z_R^2 + Z_C^2}$

where $Z_R = R$

$Z_C = \frac{1}{j\omega C}$



so, $Z_L = Z_R + Z_C$

$\frac{1}{j \cdot 2\pi \cdot 1E6} = 0.015915j\ \Omega$

$Z_L = 25 + 0.015915j\ \Omega$

$\hookrightarrow \simeq 0$

*can ignore C! (insignificant contribution)

a.) $\frac{\tilde{V}_R}{\tilde{V}_0} = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{-25}{75} = -\frac{1}{3} \simeq \Gamma$

so amplitude $\simeq \frac{1}{3}$, and phase $\simeq 180°$

b.) $\tilde{I} = \frac{V_0}{Z_0} = \frac{1}{50}$ w/ $\frac{1}{50} \cdot \frac{1}{|\Gamma|} = I = \frac{1}{50} \cdot \frac{1}{3} = \frac{3}{150} \simeq 0.667$ A

and phase $\simeq 0°$

c.) $V_R = I_L \, R$

$= \dfrac{V_L}{Z_L} \cdot R$

$= V_L = V_0(1+\Gamma)$

so, $V_R = \dfrac{V_0(1+\Gamma)}{Z_L} \cdot R = \dfrac{1-\frac{1}{3}}{25} = \dfrac{2/3}{25} \simeq 0.02\underline{66}$ A

d.) $P_{in} = \dfrac{\hat{V_0}^2}{2 Z_0} = \dfrac{1}{100}$ W, $P_{ref.} = \Gamma^2 P_{in} = \frac{1}{9} \cdot 10\text{mW} = \frac{10}{9}$ mW, $P_{res.} = \frac{1}{2} I_L \cdot R^2$

$= 10$ mW

$= \dfrac{0.0266^2 \cdot 25}{2}$

$= 0.00889$ W

$= 8.89$ mW

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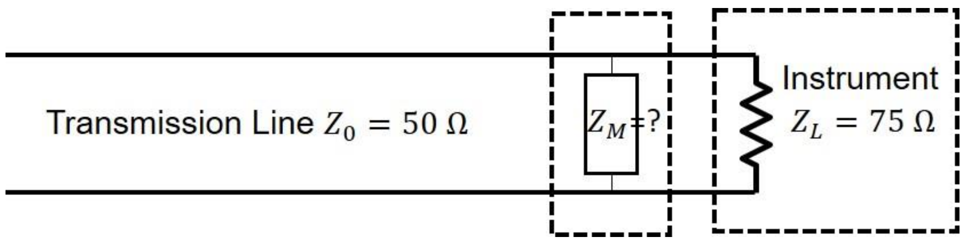

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